What are OEE, OOE, and TEEP?

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In today's industrial production environment, it is critically important for businesses to optimize the efficiency of their production processes to achieve a sustainable competitive advantage. In this optimization process, OEE (Overall Equipment Effectiveness), OOE (Overall Operations Effectiveness), and TEEP (Total Effective Equipment Performance) stand out as key metrics used in measuring and analyzing production efficiency. These three fundamental metrics help to measure production processes, identify bottlenecks, determine operational losses, and make capacity management more effective.

OEE (Overall Equipment Effectiveness) – How Efficient is Your Production Process?

OEE is a performance indicator that comprehensively evaluates the efficiency of production processes and is derived from the multiplication of three key metrics. It enables the systematic identification and analysis of losses in production processes. These three key metrics are:

- **Availability:**

  Measures how long the machine is actively operating within the planned production time.

- **Performance:**

  Indicates how efficiently the machine produces during its operating time.

Determines the percentage of defect-free products in total production.

The formula is as follows:

OEE = Availability × Performance × Quality

For example, if the planned operating time for a production line is 10 hours, but the machine has operated for only 8 hours:

Availability = 8/10 = 0.80 = 80%

If the machine has operated at 90% of its ideal speed (performance) and 95% of the produced products are defect-free (quality), then:

OEE = 0.80 × 0.90 × 0.95 = 0.684 = 68.4%

The ideal OEE value is 85% or higher. Lower values indicate areas that require improvement. However, OEE only considers the planned production time, which may not be sufficient to evaluate the entire operational process.

OOE (Overall Operations Effectiveness) – How is Actual Production Time Managed?

OEE calculation measures efficiency only within the planned production time. However, in real-world production processes, there are unplanned downtimes and capacity constraints. This is where OOE comes into play. The formula is as follows:

OOE = OEE × (Actual Production Time / Total Operation Time)

Since OOE shows how equipment is utilized throughout the entire operational time, it also accounts for unplanned downtimes and capacity constraints. For example, if a factory has a weekly operational time of 100 hours but the machines operate for only 60 hours, these losses can be measured using the OOE metric to identify areas for improvement.

OOE = OEE × 60/100

Therefore, OOE is particularly important for analyzing losses caused by unplanned maintenance processes, supply chain issues, or sudden stoppages.

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TEEP, which is used to measure the efficiency of the potential utilization time of equipment, is a more comprehensive version of OEE. This metric provides businesses with a unique perspective on the true potential of their equipment. The key difference between TEEP and OEE calculations is that TEEP is based on total calendar time.

TEEP = Loading Rate × OEE

Here, the Loading Rate is the ratio of planned production time to the total calendar time:

Loading Rate = (Planned Production Time / Total Calendar Time)

TEEP calculations consider a standard year of 365 days (8,760 hours). This approach:

- Allows for the evaluation of total equipment potential.
- Supports capacity planning processes.
- Guides investment decisions.
- Assists in shift scheduling.

For example, a factory can theoretically operate 8,760 hours per year (365 days × 24 hours). However, if the planned production time is 5,000 hours:

- Loading Rate = 5000 / 8760 = 0.57 = 57%
- If the OEE value is calculated as 70%:

TEEP = 0.57 × 0.70 = 0.399 = 39.9%

This indicates that the factory is utilizing only 39.9% of its maximum capacity. If the TEEP value is low, the business may need to review its capacity planning and optimize its production processes.

Differences Between OEE, OOE, and TEEP

OEE focuses on improving daily operations, while OOE and TEEP provide broader perspectives, contributing to long-term strategic planning.

What Is It Used For?	Measures daily production efficiency	Analyzes unplanned downtimes	Supports capacity planning
Who Uses It?	Production Engineers	Operations Managers	Senior Management and Strategy Teams

What are OEE, OOE, and TEEP?

In conclusion, for short-term improvements, OEE is the best indicator for reducing equipment failures, speed reductions, and quality issues. To analyze unplanned downtimes and prevent operational losses, OOE should be monitored. For long-term capacity management, TEEP should be evaluated to determine how much of the maximum production potential is being utilized.

Analyzing these three metrics together enables the identification of the right strategies to enhance production efficiency, minimize losses, and optimize operations.

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